

Interpreting Stable Isotope Measurements



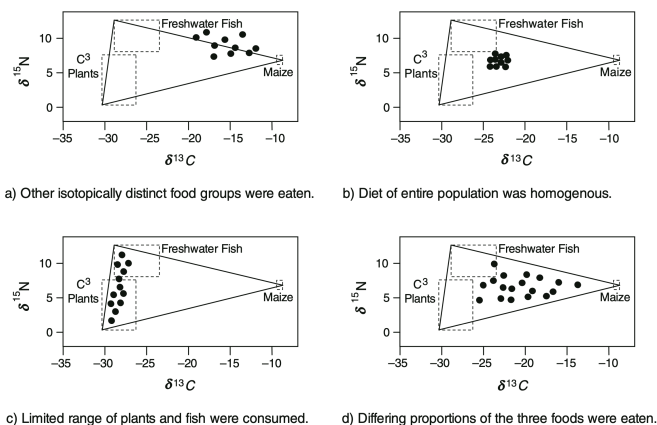
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The most challenging aspect of stable carbon and nitrogen analysis of human remains to infer diet is the interpretation of the resulting values. For example, $\delta^{13}\text{C}$ values of -20‰, intermediate between the ranges of C_3 and C_4 plants, might be the result of a diet that included equal proportions of each type of plant. But it also could be the result of a diet that included only marine C_3 plants or animals, or the result of eating only CAM plants. Combining nitrogen values with those of carbon helps to distinguish these possibilities: An individual whose diet is rich in marine animals should have a $\delta^{15}\text{N}$ value much higher than that of an individual who ate equal proportions of C_3 and C_4 plants. There are still possible confounding circumstances, however, as a low marine $\delta^{15}\text{N}$ value could be the result of eating low-trophic-level marine organisms, such as shellfish, or the result of eating a diet rich in terrestrial animal protein and C_4 plants. There is a level of equifinality inherent in most stable isotope analyses as a result!

Equifinality is the condition of two or more processes resulting in the same outcome. As a result, simply knowing the end result (such as the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of an individual) may not be sufficient to determine the process (here, the diet eaten) that would produce that outcome, as two or more processes might have produced identical outcomes.

One way of distinguishing diet looks at the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of not only one, but multiple individuals from a certain site or region. This population-level view allows various hypotheses about diet within a community or across communities to be tested. **Linear mixing models** are one tool for exploring different hypotheses about diet at the population level. In this case, such models appear as two-dimensional (X-Y) graphs that distinguish the relative concentration of three foods with distinct $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values in the diets of individuals. Each dot represents an individual tested, while dotted boxes represent the known isotope values of different potential food items. The triangle between the three food items is the potential set of values for individuals eating only these three food items. If individuals ate foods not included in the model, then their values might

fall outside of this triangle (as in the first scenario below). By looking at a population level it is possible to determine the full range of food items available to the population and the diversity among individuals, such as is shown by comparing scenarios *b* and *d* below.



Potential population-level diets of individuals eating up to three isotopically distinct foods—here, C_3 plants, freshwater fish, and maize (a C_4 plant). Note that in scenario *c*, only two foods appear to have been eaten, while in scenario *a*, at least one food not present in the model was consumed.

Source: Figure 13.2 in Malainey, M. E. (2011). *A consumer's guide to archaeological science analytical techniques*.

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Linear mixing models are mathematical and graphical models that explore the range of values possible when two or more variables are mixed. In the case of isotope studies of diet, linear mixing models are used to distinguish the input of a certain number of foods with distinct isotopic compositions in the diets of individuals; if carbon and nitrogen isotope values are modeled, the graph appears in two dimensions, and up to three distinct foods (or groups of similar foods, such as marine fish or C_4 plants) can be incorporated into the model.

Ultimately, the value of stable carbon and nitrogen isotope analysis for interpreting human diet is highest when narrowly defined questions that are suitable to isotope analysis are asked. For example, it is possible to identify when and how a new isotopically distinct food appeared in the diet, such as the arrival of the C₄ plant maize in North America, where only C₃ plants were previously eaten. Another example is exploring the age at which children were weaned (ceased exclusive consumption of breast milk), as this results in a decline in $\delta^{15}\text{N}$ values: Children cease (literally!) eating their mothers and begin to eat the same foods as their mothers, moving down a trophic level.